

# Pranathi Taddi

[pranathitaddi@gmail.com](mailto:pranathitaddi@gmail.com) | [linkedin.com/in/pranathi-taddi-662ab027b](https://linkedin.com/in/pranathi-taddi-662ab027b) | [github.com/pranathitaddi](https://github.com/pranathitaddi)

## EDUCATION

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| <b>Indian Institute of Technology Kharagpur</b><br><i>Dual Degree (B.Tech + M.Tech) Civil Engineering; Minor in Computer Science   CGPA: 8.16</i><br>• Interdisciplinary Specialization: Artificial Intelligence and Machine Learning Applications | Kharagpur, West Bengal<br>Nov 2022 – Jul 2027 |
| <b>Narayana Junior College</b><br><i>Intermediate, State Board of Telangana   Grade: 97.2%</i>                                                                                                                                                     | Hyderabad, Telangana<br>Apr 2020 – Jun 2022   |
| <b>Delhi Public School</b><br><i>CBSE Class X   Grade: 93.2%</i>                                                                                                                                                                                   | Hyderabad, Telangana<br>Aug 2017 – Apr 2020   |

## PROJECTS

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| <b>Climate ISMR Prediction Dashboard</b>   <i>Next.js, TypeScript, Python, Flask</i><br>• Developed an interactive climate analytics dashboard visualizing 50 years of Indian Summer Monsoon Rainfall (ISMR) data alongside 7 large-scale climate indices using Next.js and React Simple Maps.<br>• Designed geospatial visualizations with value-scaled markers and directed teleconnection links; integrated a Flask REST API backend enabling data-driven ISMR forecasting via a regression pipeline.                    | Feb 2026 |
| <b>DEM Parameter Extraction using ML</b>   <i>Python, ABAQUS, SVR, ANN</i>   Bachelor Thesis<br>• Built a Discrete Element Method (DEM) simulation framework in ABAQUS modeling concrete aggregate behavior via Hertz–Mindlin contact laws, parametrically varying micro-parameters across 520-particle simulations.<br>• Formulated a supervised regression pipeline mapping micro-parameters to macroscopic mechanical behavior; trained and optimized SVR, ANN, and a hybrid ANN–SVM model with k-fold cross-validation. | Nov 2025 |
| <b>Flood Early Warning System (FEWS)</b>   <i>Python, Scikit-learn, GridSearchCV</i><br>• Developed a data-driven flood risk classification system for Guwahati using preprocessed rainfall and river discharge time-series; implemented and hyperparameter-tuned Logistic Regression, Random Forest, KNN, Naive Bayes, and MLP classifiers.<br>• Achieved best performance with Random Forest at 88.19% accuracy and F1 Score of 0.57; evaluated models using F1 Score, Log Loss, and Von Mises metrics.                   | Apr 2025 |

## RELEVANT COURSEWORK

**Computer Science:** Programming and Data Structures, AI and ML Foundations and Applications, DL Foundations and Applications, Hardware Security, Computer Networks, Switching Circuits and Logic Design, ML in WRE  
**Mathematics:** Probability and Statistics, Functional Analysis, Linear Algebra for AI/ML, Numerical and Complex Analysis, Advanced Calculus

## TECHNICAL SKILLS

**Languages:** Python, C++, C, JavaScript, TypeScript, Bash, HTML/CSS, Java  
**Frameworks & Libraries:** Next.js, Node.js, Express.js, Redux, Flask, Django, Angular, PyTorch, TensorFlow, Keras, Spring Boot  
**Developer Tools:** Git, GitHub, VS Code, Jupyter Notebook, Linux/Unix, MongoDB, Firebase, Tailwind CSS, ShadCN, ABAQUS

## LEADERSHIP & EXPERIENCE

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| <b>Web Team Coordinator</b><br><i>TEDxIITKharagpur, Indian Institute of Technology Kharagpur</i><br>• Led the full relaunch of the official TEDxIITKharagpur website using Angular and Next.js with Tailwind CSS, improving performance, visual design, and compliance with TED global branding guidelines.<br>• Managed a team of 5+ developers to deliver responsive UI components, interactive gallery modules, and structured event navigation features on schedule. | Jul 2024 – Present  |
| <b>Technical Team Sub-Head</b><br><i>Kshitij, Indian Institute of Technology Kharagpur</i><br>• Collaborated with a 15+ member team to build Kshitij'23, a 3D interactive techno-management festival website using Three.js and the MERN stack, attracting 7,000+ visitors.<br>• Co-organized large-scale technical events including an overnight coding contest with thousands of participants across India.                                                            | Aug 2023 – Jan 2024 |